

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition? Answer Choices: A. It activates TLR5 on the surface of macrophages. B. It primarily induces the Th1-cell response. C. Peptidoglycan is its major antigen that induces an immune response. D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient's condition is community-acquired pneumonia that is treated with cephalexin, which is an antibiotic that is effective against gram-positive bacteria. The patient's symptoms and the fact that cephalexin provided modest improvement suggest that the cause of the pneumonia is likely a gram-positive bacterium. The immune response elicited by gram-positive bacteria involves the activation of the innate immune response, which recognizes peptidoglycan as a major antigen. Peptidoglycan is a component of the bacterial cell wall and is recognized by the immune system as a pathogen-associated molecular pattern (PAMP). This leads to the activation of the innate immune response, including the activation of macrophages and the production of cytokines. The Th1-cell response is more associated with intracellular pathogens, such as viruses and some bacteria that evade the immune response by encryption like Mycobacterium tuberculosis. TLR5 is activated by bacterial flagellin, which is a component of gram-negative bacteria. Therefore, the correct answer is the one that mentions peptidoglycan as the major antigen that induces an immune response. </think> <answer> C . Peptidoglycan is its major antigen that induces an immune response. </answer>

